Anuprita Mhapankar **D20A** **34**

Blockchain Experiment 9

AIM: Implement the Private Ethereum Blockchain using Geth

THEORY:

Q1: What is Geth?

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through JSON-RPC, command-line, or JavaScript consoles.

Q2: Significance of a Private Ethereum Network

Setting up a Private Ethereum Network involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

Q3: Steps for creating a Private Ethereum Network

Steps Involved:

- **Choosing a Network ID:**
  Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
- **Choosing a Consensus Algorithm:**
  Ethereum supports different consensus mechanisms such as Proof of Work (PoW) and Proof of Authority (PoA). In private networks, Clique (PoA) is often used for faster block confirmation.
- **Creating a Genesis Block:**
  The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
- **Initializing the Geth Database:**
  The Geth init command initializes the blockchain using the genesis.json file, preparing the data directory for node operation.
- **Setting up Networking:**
  Nodes communicate using enodes (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
- **Running the Member Nodes:**
  Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
- **Running a Signer (in Clique):**
  In the Clique consensus, designated signers validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

NOTE: Download the genesis file and edit the account details, including the public keys of all participating peers in the network.

PROCEDURE

Step 1: Install Geth

wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
mkdir -p ~/bin
cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
export PATH="$HOME/bin:$PATH"
geth113 version

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
g2022_anuprita_mhapankar@cloudshell:~$ wget https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
--2026-04-05 12:55:59--  https://gethstore.blob.core.windows.net/builds/geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
Resolving gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)... 20.60.40.164
Connecting to gethstore.blob.core.windows.net (gethstore.blob.core.windows.net)|20.60.40.164|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 28248613 (27M) [application/octet-stream]
Saving to: 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz'

geth-linux-amd64-1.13.14-2bd6bd01.tar.gz    100%[===================================================================================>]  26.94M  4.79MB/s

2026-04-05 12:56:08 (3.47 MB/s) - 'geth-linux-amd64-1.13.14-2bd6bd01.tar.gz' saved [28248613/28248613]

g2022_anuprita_mhapankar@cloudshell:~$ tar -xvf geth-linux-amd64-1.13.14-2bd6bd01.tar.gz
geth-linux-amd64-1.13.14-2bd6bd01/
geth-linux-amd64-1.13.14-2bd6bd01/COPYING
geth-linux-amd64-1.13.14-2bd6bd01/geth
g2022_anuprita_mhapankar@cloudshell:~$ mkdir -p ~/bin
g2022_anuprita_mhapankar@cloudshell:~$ cp geth-linux-amd64-1.13.14-2bd6bd01/geth ~/bin/geth113
g2022_anuprita_mhapankar@cloudshell:~$ export PATH="$HOME/bin:$PATH"
g2022_anuprita_mhapankar@cloudshell:~$ geth113 version
Geth
Version: 1.13.14-stable
Git Commit: 2bd6bd01d2e8561dd7fc21b631f4a34ac16627a1
Git Commit Date: 20240227
Architecture: amd64
Go Version: go1.21.6
Operating System: linux
GOPATH=/home/g2022_anuprita_mhapankar/gopath:/google/gopath
GOROOT=
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 2: Create folders and accounts for both nodes

```
g2022_anuprita_mhapankar@cloudshell:~$ mkdir -p ~/eth-private/node1 ~/eth-private/node2
g2022_anuprita_mhapankar@cloudshell:~$ geth113 --datadir ~/eth-private/node1 account new
INFO [04-05|12:58:38.464] Maximum peer count                       ETH=50 total=50
INFO [04-05|12:58:38.467] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x539C7Cfe00fa9E226fe522593400a9BC9383a317
Path of the secret key file: /home/g2022_anuprita_mhapankar/eth-private/node1/keystore/UTC--2026-04-05T12-58-44.498199756Z--539c7cfe00fa9e226fe522593400a9bc9383a317

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

g2022_anuprita_mhapankar@cloudshell:~$
```

```
g2022_anuprita_mhapankar@cloudshell:~$ geth113 --datadir ~/eth-private/node2 account new
INFO [04-05|12:59:28.005] Maximum peer count                       ETH=50 total=50
INFO [04-05|12:59:28.009] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
Your new account is locked with a password. Please give a password. Do not forget this password.
Password:
Repeat password:

Your new key was generated

Public address of the key:   0x5fd63344c1f40712A9A85d9d7e2B304a6C558716
Path of the secret key file: /home/g2022_anuprita_mhapankar/eth-private/node2/keystore/UTC--2026-04-05T12-59-32.919304438Z--5fd63344c1f40712a9a85d9d7e2b304a6c558716

- You can share your public address with anyone. Others need it to interact with you.
- You must NEVER share the secret key with anyone! The key controls access to your funds!
- You must BACKUP your key file! Without the key, it's impossible to access account funds!
- You must REMEMBER your password! Without the password, it's impossible to decrypt the key!

g2022_anuprita_mhapankar@cloudshell:~$
```

Step 3: Create password files

```
g2022_anuprita_mhapankar@cloudshell:~$ echo "1234" > ~/eth-private/node1/password.txt
g2022_anuprita_mhapankar@cloudshell:~$ echo "1234" > ~/eth-private/node2/password.txt
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 4: Create genesis.json and verify it looks correct

```
NODE1="PASTE_NODE1_ADDRESS_WITHOUT_0x_HERE"
NODE2="PASTE_NODE2_ADDRESS_WITHOUT_0x_HERE"

cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata":
"0x0000000000000000000000000000000000000000000000000000000000000000${NODE1
}0000000000000000000000000000000000000000000000000000000000000000000000000
0000000000000000000000000000000000000000000000000000000000000000",
  "alloc": {
    "0x${NODE1}": { "balance": "10000000000000000000" },
    "0x${NODE2}": { "balance": "10000000000000000000" }
  }
}
EOF
```

```
g2022_anuprita_mhapankar@cloudshell:~$ NODE1="539C7Cfe00fa9E226fe522593400a9BC9383a317"
NODE2="5fd63344c1f40712A9A85d9d7e2B304a6C558716"

cat > ~/eth-private/genesis.json << EOF
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata": "0x0000000000000000000000000000000000000000000000000000000000000000${NODE1}0000000000000000000000000000
0000000000000000000000000000000000000000000000",
  "alloc": {
    "0x${NODE1}": { "balance": "10000000000000000000" },
    "0x${NODE2}": { "balance": "10000000000000000000" }
  }
}
EOF
g2022_anuprita_mhapankar@cloudshell:~$
```

```
g2022_anuprita_mhapankar@cloudshell:~$ cat ~/eth-private/genesis.json
{
  "config": {
    "chainId": 1234,
    "homesteadBlock": 0,
    "eip150Block": 0,
    "eip155Block": 0,
    "eip158Block": 0,
    "byzantiumBlock": 0,
    "constantinopleBlock": 0,
    "petersburgBlock": 0,
    "clique": {
      "period": 5,
      "epoch": 30000
    }
  },
  "difficulty": "1",
  "gasLimit": "8000000",
  "extradata": "0x0000000000000000000000000000000000000000000000000000000000000000539C7Cfe00fa9E226fe522593400a9BC9383a31700000000000
0000000000000000000000000000000000000000000000000000000000000000000000000000",
  "alloc": {
    "0x539C7Cfe00fa9E226fe522593400a9BC9383a317": { "balance": "10000000000000000000" },
    "0x5fd63344c1f40712A9A85d9d7e2B304a6C558716": { "balance": "10000000000000000000" }
  }
}
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 5: Initialize both nodes

```
g2022_anuprita_mhapankar@cloudshell:~$ geth113 --datadir ~/eth-private/node1 init ~/eth-private/genesis.json
INFO [04-05|13:07:13.568] Maximum peer count                       ETH=50 total=50
INFO [04-05|13:07:13.571] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-05|13:07:13.578] Set global gas cap                       cap=50,000,000
INFO [04-05|13:07:13.579] Initializing the KZG library             backend=gokzg
INFO [04-05|13:07:13.714] Defaulting to pebble as the backing database
INFO [04-05|13:07:13.714] Allocated cache and file handles         database=/home/g2022_anuprita_mhapankar/eth-private/node1/geth/chaindata cache=16.00MiB handles=16
INFO [04-05|13:07:13.789] Opened ancient database                  database=/home/g2022_anuprita_mhapankar/eth-private/node1/geth/chaindata/ancient/chain readonly=false
INFO [04-05|13:07:13.789] State schema set to default              scheme=hash
INFO [04-05|13:07:13.789] Writing custom genesis block
INFO [04-05|13:07:13.794] Persisted trie from memory database      nodes=3 size=411.00B time=4.479841ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-05|13:07:13.842] Successfully wrote genesis state         database=chaindata hash=17443b..43ee61
INFO [04-05|13:07:13.843] Defaulting to pebble as the backing database
INFO [04-05|13:07:13.843] Allocated cache and file handles         database=/home/g2022_anuprita_mhapankar/eth-private/node1/geth/lightchaindata cache=16.00MiB handles=16
INFO [04-05|13:07:13.919] Opened ancient database                  database=/home/g2022_anuprita_mhapankar/eth-private/node1/geth/lightchaindata/ancient/chain readonly=false
INFO [04-05|13:07:13.919] State schema set to default              scheme=hash
INFO [04-05|13:07:13.919] Writing custom genesis block
INFO [04-05|13:07:13.924] Persisted trie from memory database      nodes=3 size=411.00B time=4.415771ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-05|13:07:13.971] Successfully wrote genesis state         database=lightchaindata hash=17443b..43ee61
g2022_anuprita_mhapankar@cloudshell:~$
```

```
g2022_anuprita_mhapankar@cloudshell:~$ geth113 --datadir ~/eth-private/node2 init ~/eth-private/genesis.json
INFO [04-05|13:07:50.761] Maximum peer count                       ETH=50 total=50
INFO [04-05|13:07:50.766] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-05|13:07:50.771] Set global gas cap                       cap=50,000,000
INFO [04-05|13:07:50.772] Initializing the KZG library             backend=gokzg
INFO [04-05|13:07:50.910] Defaulting to pebble as the backing database
INFO [04-05|13:07:50.910] Allocated cache and file handles         database=/home/g2022_anuprita_mhapankar/eth-private/node2/geth/chaindata cache=16.00MiB handles=16
INFO [04-05|13:07:50.986] Opened ancient database                  database=/home/g2022_anuprita_mhapankar/eth-private/node2/geth/chaindata/ancient/chain readonly=false
INFO [04-05|13:07:50.986] State schema set to default              scheme=hash
INFO [04-05|13:07:50.986] Writing custom genesis block
INFO [04-05|13:07:50.992] Persisted trie from memory database      nodes=3 size=411.00B time=5.821691ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-05|13:07:51.043] Successfully wrote genesis state         database=chaindata hash=17443b..43ee61
INFO [04-05|13:07:51.044] Defaulting to pebble as the backing database
INFO [04-05|13:07:51.044] Allocated cache and file handles         database=/home/g2022_anuprita_mhapankar/eth-private/node2/geth/lightchaindata cache=16.00MiB handles=16
INFO [04-05|13:07:51.120] Opened ancient database                  database=/home/g2022_anuprita_mhapankar/eth-private/node2/geth/lightchaindata/ancient/chain readonly=false
INFO [04-05|13:07:51.120] State schema set to default              scheme=hash
INFO [04-05|13:07:51.120] Writing custom genesis block
INFO [04-05|13:07:51.126] Persisted trie from memory database      nodes=3 size=411.00B time=4.735292ms gcnodes=0 gcsize=0.00B gctime=0s livenodes=0 livesize=0.00B
INFO [04-05|13:07:51.175] Successfully wrote genesis state         database=lightchaindata hash=17443b..43ee61
g2022_anuprita_mhapankar@cloudshell:~$
```

Step 6: Run Node 1 (signer)

```
g2022_anuprita_mhapankar@cloudshell:~$ geth113 --datadir ~/eth-private/node1 \
  --networkid 1234 \
  --port 30303 \
  --http --http.port 8545 \
  --unlock 0x${NODE1} \
  --password ~/eth-private/node1/password.txt \
  --mine \
  --miner.etherbase 0x${NODE1} \
  --allow-insecure-unlock \
  console
INFO [04-05|13:09:33.554] Maximum peer count                       ETH=50 total=50
INFO [04-05|13:09:33.558] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-05|13:09:33.563] Set global gas cap                       cap=50,000,000
INFO [04-05|13:09:33.564] Initializing the KZG library             backend=gokzg
INFO [04-05|13:09:33.703] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-05|13:09:33.704] Using pebble as the backing database
INFO [04-05|13:09:33.704] Allocated cache and file handles         database=/home/g2022_anuprita_mhapankar/eth-private/node1/geth/chaindata cache=512.00MiB handles=524,288
INFO [04-05|13:09:33.743] Opened ancient database                  database=/home/g2022_anuprita_mhapankar/eth-private/node1/geth/chaindata/ancient/chain readonly=false
INFO [04-05|13:09:33.743] State scheme set to already existing     scheme=hash
INFO [04-05|13:09:33.744] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [04-05|13:09:33.745]
INFO [04-05|13:09:33.745] ----------------------------------------------------------------------------------------------------------------------------------------
----
INFO [04-05|13:09:33.745] Chain ID:  1234 (unknown)
INFO [04-05|13:09:33.745] Consensus: Clique (proof-of-authority)
INFO [04-05|13:09:33.745]
INFO [04-05|13:09:33.745] Pre-Merge hard forks (block based):
INFO [04-05|13:09:33.745]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [04-05|13:09:33.745]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistl
e.md)
INFO [04-05|13:09:33.745]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.
md)
INFO [04-05|13:09:33.745]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.
md)
INFO [04-05|13:09:33.745]  - Byzantium:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/byzantium.md)
INFO [04-05|13:09:33.745]  - Constantinople:               #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/constantinople.m
d)
INFO [04-05|13:09:33.745]  - Petersburg:                   #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/petersburg.md)
INFO [04-05|13:09:33.745]  - Istanbul:                     #<nil> (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/istanbul.md)
```

```
INFO [04-05|13:09:33.851] WebSocket enabled                        url=ws://127.0.0.1:8551
INFO [04-05|13:09:33.851] HTTP server started                      endpoint=127.0.0.1:8551 auth=true  prefix= cors=localhost vhosts=localhost
INFO [04-05|13:09:35.182] Unlocked account                         address=0x539C7Cfe00fa9E226fe522593400a9BC9383a317
INFO [04-05|13:09:35.182] Legacy pool tip threshold updated        tip=0
INFO [04-05|13:09:35.182] Legacy pool tip threshold updated        tip=1,000,000,000
INFO [04-05|13:09:35.182] Snap sync complete, auto disabling
INFO [04-05|13:09:35.195] Commit new sealing work                  number=1 sealhash=e9e9b6..788de6 txs=0 gas=0 fees=0 elapsed="188.08µs"
INFO [04-05|13:09:35.206] Successfully sealed new block            number=1 sealhash=e9e9b6..788de6 hash=c785c9..acccc0 elapsed=10.150ms
INFO [04-05|13:09:35.207] Commit new sealing work                  number=2 sealhash=e6a32e..e52088 txs=0 gas=0 fees=0 elapsed="768.316µs"
INFO [04-05|13:09:35.213] Indexed transactions                     blocks=2 txs=0 tail=0 elapsed=4.540ms
Welcome to the Geth JavaScript console!

instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
coinbase: 0x539c7cfe00fa9e226fe522593400a9bc9383a317
at block: 1 (Sun Apr 05 2026 13:09:35 GMT+0000 (UTC))
 datadir: /home/g2022_anuprita_mhapankar/eth-private/node1
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-05|13:09:36.325] New local node record                    seq=1,775,394,512,621 id=fe815edca75775d6 ip=34.143.138.199 udp=30303 tcp=30303
INFO [04-05|13:09:40.016] Successfully sealed new block            number=2 sealhash=e6a32e..e52088 hash=13bda8..0a85a0 elapsed=4.809s
INFO [04-05|13:09:40.017] Commit new sealing work                  number=3 sealhash=ef86a8..40eec6 txs=0 gas=0 fees=0 elapsed=1.369ms
INFO [04-05|13:09:44.599] Looking for peers                        peercount=0 tried=121 static=0
INFO [04-05|13:09:45.028] Successfully sealed new block            number=3 sealhash=ef86a8..40eec6 hash=cda713..a6ccf8 elapsed=5.010s
INFO [04-05|13:09:45.029] Commit new sealing work                  number=4 sealhash=d5303a..7b8175 txs=0 gas=0 fees=0 elapsed="653.519µs"
INFO [04-05|13:09:50.013] Successfully sealed new block            number=4 sealhash=d5303a..7b8175 hash=69e616..c2c61f elapsed=4.983s
INFO [04-05|13:09:50.014] Commit new sealing work                  number=5 sealhash=39ae2c..872540 txs=0 gas=0 fees=0 elapsed="503.517µs"
INFO [04-05|13:09:54.807] Looking for peers                        peercount=0 tried=40  static=0
INFO [04-05|13:09:55.015] Successfully sealed new block            number=5 sealhash=39ae2c..872540 hash=a1fbd7..8ebe44 elapsed=5.001s
INFO [04-05|13:09:55.016] Commit new sealing work                  number=6 sealhash=87e0fe..b886a7 txs=0 gas=0 fees=0 elapsed="699.073µs"
INFO [04-05|13:10:00.012] Successfully sealed new block            number=6 sealhash=87e0fe..b886a7 hash=6585f4..dfc3c6 elapsed=4.995s
```

Step 7: Open new terminal tab and run Node 2 (member)

```
g2022_anuprita_mhapankar@cloudshell:~$ export PATH="$HOME/bin:$PATH"
NODE2="5fd63344c1f40712A9A85d9d7e2B304a6C558716"

geth113 --datadir ~/eth-private/node2 \
  --networkid 1234 \
  --port 30304 \
  --http --http.port 8546 \
  --authrpc.port 8552 \
  --unlock 0x${NODE2} \
  --password ~/eth-private/node2/password.txt \
  --allow-insecure-unlock \
  console
INFO [04-05|13:16:40.494] Maximum peer count                       ETH=50 total=50
INFO [04-05|13:16:40.498] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such file or directory"
INFO [04-05|13:16:40.503] Set global gas cap                       cap=50,000,000
INFO [04-05|13:16:40.504] Initializing the KZG library             backend=gokzg
INFO [04-05|13:16:40.657] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [04-05|13:16:40.657] Using pebble as the backing database
INFO [04-05|13:16:40.657] Allocated cache and file handles         database=/home/g2022_anuprita_mhapankar/eth-private/node2/geth/chaindata cache=512.00MiB handles=524,288
INFO [04-05|13:16:40.696] Opened ancient database                  database=/home/g2022_anuprita_mhapankar/eth-private/node2/geth/chaindata/ancient/chain readonly=false
INFO [04-05|13:16:40.697] State scheme set to already existing     scheme=hash
INFO [04-05|13:16:40.698] Initialising Ethereum protocol           network=1234 dbversion=8
INFO [04-05|13:16:40.699]
INFO [04-05|13:16:40.699] ----------------------------------------------------------------------------------------------------------------------------------------------------------
----
INFO [04-05|13:16:40.699] Chain ID:  1234 (unknown)
INFO [04-05|13:16:40.699] Consensus: Clique (proof-of-authority)
INFO [04-05|13:16:40.699]
INFO [04-05|13:16:40.699] Pre-Merge hard forks (block based):
INFO [04-05|13:16:40.699]  - Homestead:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/homestead.md)
INFO [04-05|13:16:40.699]  - Tangerine Whistle (EIP 150): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/tangerine-whistl
e.md)
INFO [04-05|13:16:40.699]  - Spurious Dragon/1 (EIP 155): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.
md)
INFO [04-05|13:16:40.699]  - Spurious Dragon/2 (EIP 158): #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/spurious-dragon.
md)
INFO [04-05|13:16:40.699]  - Byzantium:                    #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/byzantium.md)
INFO [04-05|13:16:40.699]  - Constantinople:               #0        (https://github.com/ethereum/execution-specs/blob/master/network-upgrades/mainnet-upgrades/constantinople.m
d)
```

```
INFO [04-05|13:16:40.850] WebSocket enabled                        url=ws://127.0.0.1:8552
INFO [04-05|13:16:40.850] HTTP server started                      endpoint=127.0.0.1:8552 auth=true  prefix= cors=localhost vhosts=localhost
INFO [04-05|13:16:42.636] Unlocked account                         address=0x5fd63344c1f40712A9A85d9d7e2B304a6C558716
WARN [04-05|13:16:42.698] Served eth_coinbase                      reqid=3 duration="24.608µs" err="etherbase must be explicitly specified"
Welcome to the Geth JavaScript console!

instance: Geth/v1.13.14-stable-2bd6bd01/linux-amd64/go1.21.6
at block: 0 (Thu Jan 01 1970 00:00:00 GMT+0000 (UTC))
 datadir: /home/g2022_anuprita_mhapankar/eth-private/node2
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> INFO [04-05|13:16:43.591] New local node record                    seq=1,775,394,751,484 id=7504a412f0522db8 ip=34.143.138.199 udp=30304 tcp=30304
INFO [04-05|13:16:50.991] Looking for peers                        peercount=0 tried=38 static=0
INFO [04-05|13:17:01.133] Looking for peers                        peercount=0 tried=41 static=0
INFO [04-05|13:17:11.176] Looking for peers                        peercount=0 tried=15 static=0
INFO [04-05|13:17:21.201] Looking for peers                        peercount=0 tried=69 static=0
INFO [04-05|13:17:31.368] Looking for peers                        peercount=0 tried=40 static=0
INFO [04-05|13:17:41.668] Looking for peers                        peercount=0 tried=35 static=0
```

Step 8: Connect the two nodes

In Node 1 console, get its enode

```
> admin.nodeInfo.enode
"enode://9616b481555174819bff02f6543f7a421801c1e895b832024002b9bbebb150fbf2893fd0c09d33492303daa71d7893777c02d2a1e88ce6b425a3a72fe6165596@34.143.138.199:30303"
```

Now, switch to Node 2 console and paste the Node 1 enode

```
> admin.addPeer("enode://9616b481555174819bff02f6543f7a421801c1e895b832024002b9bbebb150fbf2893fd0c09d33492303daa71d7893777c02d2a1e88ce6b425a3a72fe6165596@34.143.138.199:30303"
)
true
```

In Node 2 console, check they are connected

```
> admin.peers
[{
    caps: ["eth/68", "eth/69", "snap/1"],
    enode: "enode://00162513d2f3e10b7ccd0dc4a3d139842e3f9681147bf7367aa7fe40421ce3904f8a4ea0db1254cc2d0d8a0dfec08457c3938017bf424a952c92c70931183abb@47.187.149.99:31404",
    id: "26184e724beb8194376e59f1fd0493aee8d3cb40469857695c966147c8b8aeeb",
    name: "Nethermind/v1.36.0+31cb81b7/linux-x64/dotnet10.0.1",
    network: {
      inbound: false,
      localAddress: "10.88.0.4:55586",
      remoteAddress: "47.187.149.99:31404",
      static: false,
      trusted: false
    },
    protocols: {
      eth: "handshake",
      snap: "handshake"
    }
}]
```

Step 9: Send a transaction

In Node 1 console, send transaction to Node 2

```
> eth.sendTransaction({
......    from: eth.accounts[0],
......    to: "0x5fd63344c1f40712A9A85d9d7e2B304a6C558716",
......    value: web3.toWei(1, "ether")
...... })
INFO [04-05|13:43:48.276] Setting new local account                address=0x1D8b6d4F6C677D2fB66CA97FbD81bED2cD23c4D3
INFO [04-05|13:43:48.276] Submitted transaction                    hash=0x566ac1b0f280ebd7b6d4a68991416a36248ed3ec43bd24f8a08af6badab79089 from=0x1D8b6d4F6C677D2fB66CA97FbD81b
ED2cD23c4D3 nonce=0 recipient=0x5fd63344c1f40712A9A85d9d7e2B304a6C558716 value=1,000,000,000,000,000,000
"0x566ac1b0f280ebd7b6d4a68991416a36248ed3ec43bd24f8a08af6badab79089"
```

and check the balance of Node 2

```
> eth.getBalance("0x5fd63344c1f40712A9A85d9d7e2B304a6C558716")
10100000000000000000
```

CONCLUSION:

Through this experiment, two smart contracts - OrderRecord and ReviewVerification, were successfully created and deployed on the Ethereum Sepolia test network using Solidity and Remix IDE. The experiment demonstrated how blockchain technology can be integrated into a full-stack web application to permanently and transparently record order transactions and product reviews. The use of the Sepolia test network and MetaMask allowed safe testing without real funds. Overall, this experiment provided practical understanding of designing, compiling, deploying, and testing smart contracts on the Ethereum blockchain.